

LSM Week 1 - R assignment

W1Q1a

"

Simulate 100 samples from Poisson distribution with parameter $\lambda = 2$ by using command `rpois(100, lambda)` in R-software and store it in vector `x`.

"

```
x <- rpois(100, lambda = 2)
```

x

```
r$> # W1Q1a
"
Simulate 100 samples from Poisson distribution with parameter  $\lambda = 2$  by using
command rpois(100, lambda) in R-software and store it in vector x.
"
x <- rpois(100, lambda = 2)
x
[1] "\nSimulate 100 samples from Poisson distribution with parameter  $\lambda = 2$  by using\ncommand rpois(100, lambda) in R-software and store it in vector x.\n"
[1] 2 3 1 3 2 2 2 3 6 2 1 5 2 3 2 1 1 1 1 1 0 3 0 0 1 1 5 1 2 3 2 0 3 2 2 1 1 2 2 2 2 1 1 5 2 4 4 5 1 3 3 2
[53] 5 2 2 2 1 3 0 4 4 1 3 4 3 6 4 5 2 0 0 1 3 2 3 2 1 1 2 2 3 5 4 2 2 4 1 2 4 0 4 3 2 3 2 4 3 0 2 1
```

W1Q1b

"

Find the summary of the generated dataset by using command `summary()` in R-software.

"

```
summary(x)
```

```
r$> # W1Q1b
"
Find the summary of the generated dataset by using
command summary() in R-software.
"
summary(x)
[1] "\nFind the summary of the generated dataset by using\ncommand summary() in R-software.\n"
Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00 1.00 2.00 2.29 3.00 6.00
```

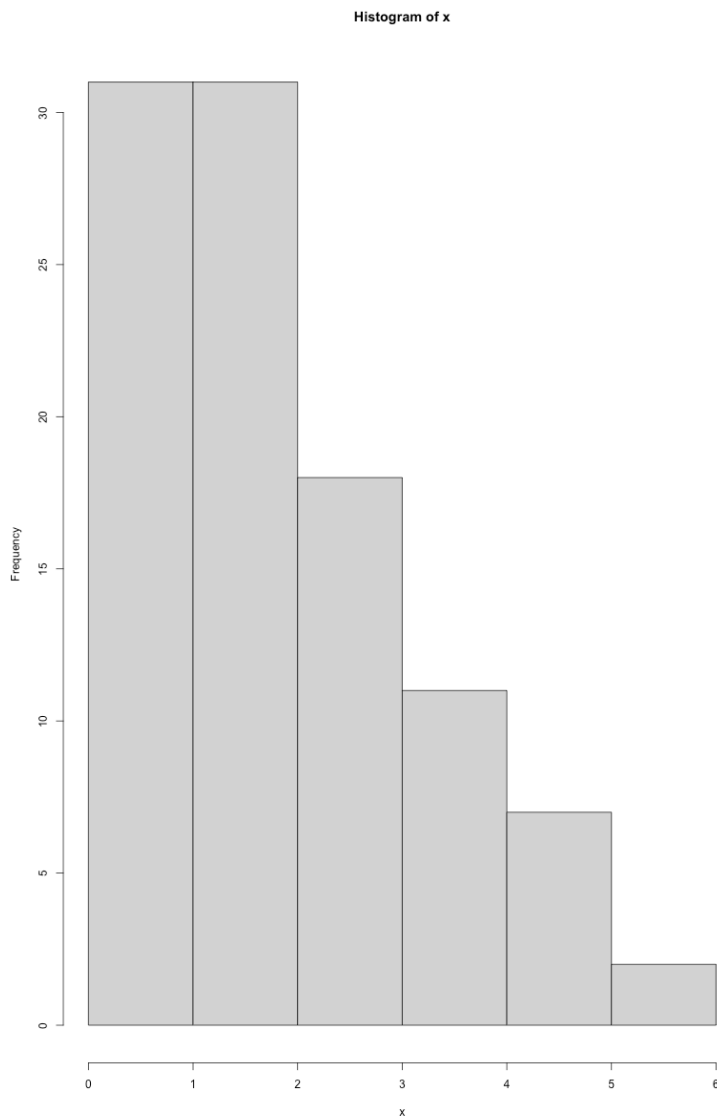
W1Q1c

"

Plot the histogram of the generated dataset the by using the command `hist()` in R-software.

"

```
hist(x)
```



W1Q2a

"

Simulate 1000 samples from Normal distribution with parameters $\mu = 20$ and $\sigma^2 = 16$ by using command `rnorm(1000,  $\mu$ ,  $\sigma$ )` in R-software and store it in vector

y.

"

`y <- rnorm(1000, mean = 20, sd = sqrt(16))`

y

```
r$> # W1Q2a
"
  Simulate 1000 samples from Normal distribution with parameters  $\mu = 20$  and
 $\sigma^2 = 16$  by using command rnorm(1000,  $\mu$ ,  $\sigma$ ) in R-software and store it in vector
y.
"
y <- rnorm(1000, mean = 20, sd = sqrt(16))
y
[1] "\nSimulate 1000 samples from Normal distribution with parameters  $\mu = 20$  and\n $\sigma^2 = 16$  by using command r
norm(1000,  $\mu$ ,  $\sigma$ ) in R-software and store it in vector\ny.\n"
[1] 14.187307 19.657654 24.349109 19.057605 27.435011 24.572317 14.518290 20.321532 22.837310 17.985719
[11] 21.019628 26.312688 20.398148 19.717645 18.722457 21.666652 29.653058 19.271290 23.619599 22.871242
[21] 18.373090 23.195847 22.401361 15.999660 24.129775 22.269754 21.174660 19.882810 16.645984 18.309087
[31] 22.928703 20.632074 20.417134 24.236454 19.429831 20.712694 21.421379 16.606444 24.850532 21.920974
[41] 17.269458 17.013799 18.355341 22.801177 18.630680 17.200244 21.168929 20.128149 19.007909 20.904579
```

```
# W1Q2b
```

```
"
```

Find the summary of the generated dataset by using command `summary()` in R-software.

```
"
```

```
summary(y)
```

```
r$> # W1Q2b
"
Find the summary of the generated dataset by using command summary() in R-
software.
"
summary(y)
[1] "\nFind the summary of the generated dataset by using command summary() in R-\nsoftware.\n"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 7.731 17.340  20.124  20.070  22.740  34.586
```

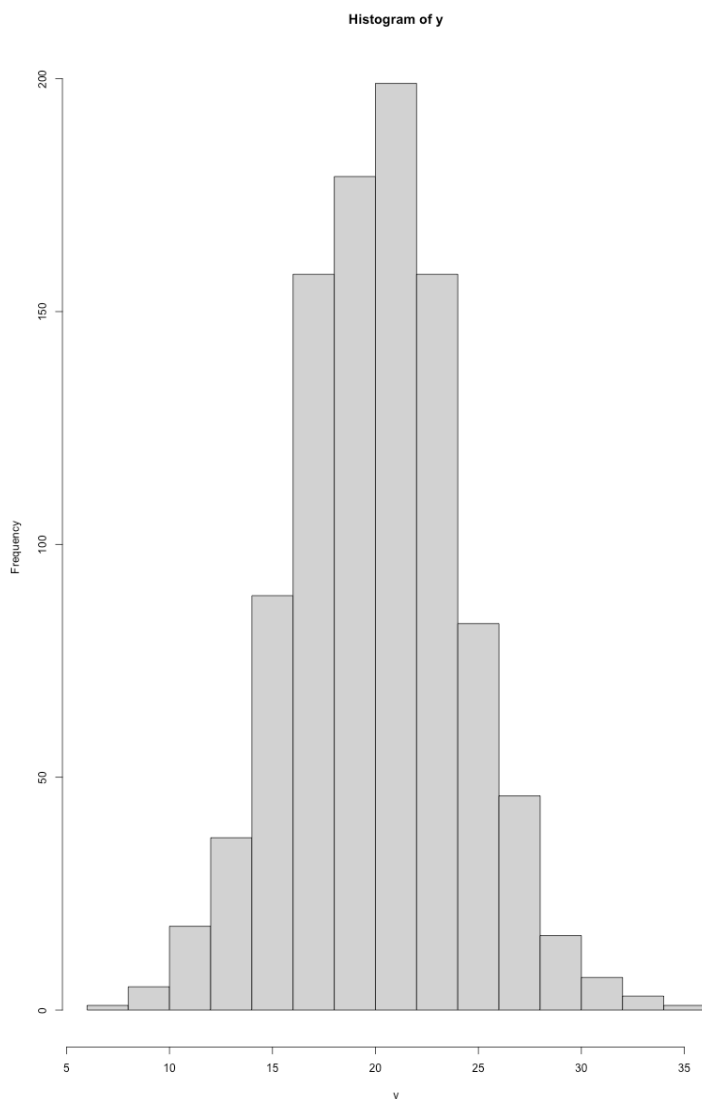
```
# W1Q2c
```

```
"
```

Plot the histogram of the generated dataset the by using the command `hist()` in R-software.

```
"
```

```
hist(y)
```



W1Q3a

"

Simulate 50 samples from continuous uniform distribution with parameters $b = 40$ and $a = 20$ by using the command `runif(50, a, b)` in R-software and store it in vector `z`.

"

```
z <- runif(50, min = 20, max = 40)
```

`z`

```
r$> # W1Q3a
"
Simulate 50 samples from continuous uniform distribution with parameters b = 40
and a = 20 by using the command runif(50, a, b) in R-software and store it in
vector z.
"
z <- runif(50, min = 20, max = 40)
z
[1] "\nSimulate 50 samples from continuous uniform distribution with parameters b = 40\nand a = 20 by using t
he command runif(50, a, b) in R-software and store it in\nvector z.\n"
[1] 22.47288 20.53063 27.32478 30.75191 29.58248 23.23451 21.17238 31.49040 39.16878 20.93934 22.84848
[12] 39.96094 22.61846 34.41105 27.33644 31.12914 32.17013 20.29977 39.62562 32.50087 33.78226 28.52753
[23] 36.73715 35.56553 37.24553 31.56942 29.63177 30.28004 37.03084 28.44646 24.17286 23.93723 24.97776
[34] 35.48231 24.16334 32.00819 32.60308 36.67985 34.12822 26.78581 39.43197 29.81584 32.21351 27.51974
[45] 37.08727 37.33476 23.82393 25.63983 32.44389 20.83094
```

W1Q3b

"

Find the summary of the generated dataset by using command `summary()` in R-software.

"

```
summary(z)
```

```
r$> # W1Q3b
"
Find the summary of the generated dataset by using command summary() in R-
software.
"
summary(z)
[1] "\nFind the summary of the generated dataset by using command summary() in R-\nsoftware.\n"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 20.30  24.37   30.52   29.99  34.34   39.96
```

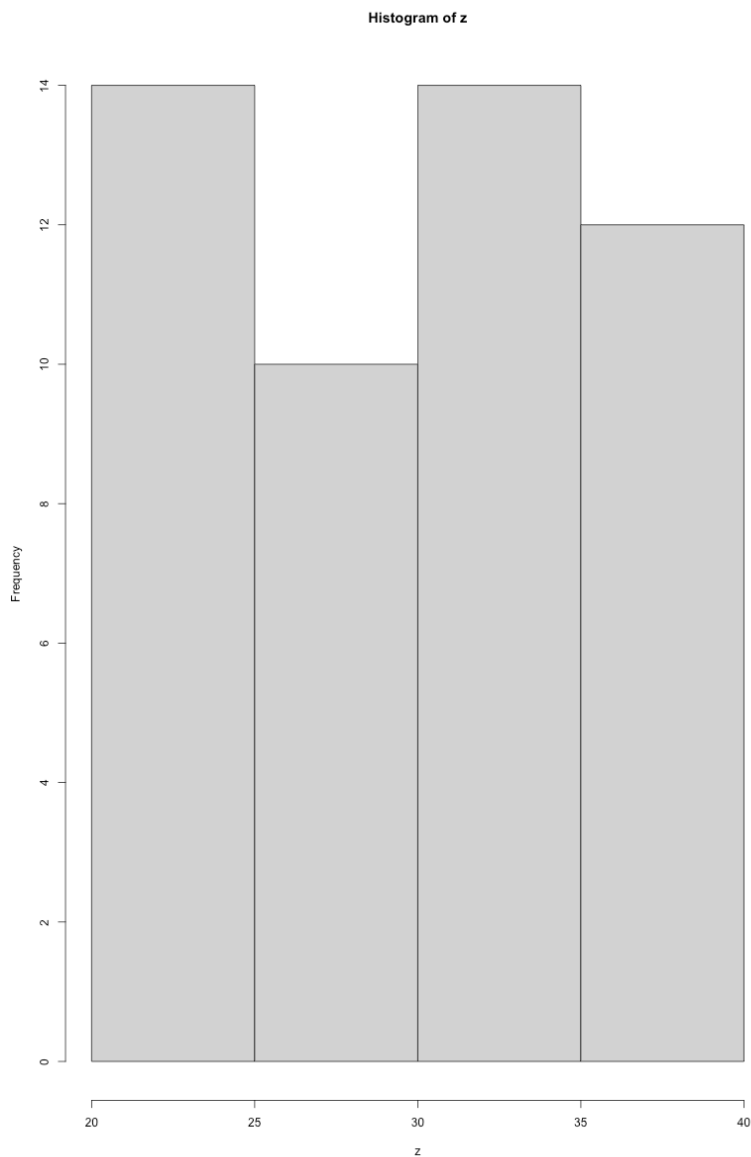
W1Q3c

"

Plot the histogram of the generated dataset the by using the command `hist()` in R-software.

"

```
hist(z)
```



W1Q4

"

An analyst wishes to study inheritance of traits from generation to generation. For this, he collected the data (given in the file heights.txt) of mothers' height (Mheight) and the height of one of their adult daughter (Dheight). Based on the given information, answer the following:
Note: Explore and try out different commands in R for computation. Also, the plots should be properly labelled

"

W1Q4a

"

Read the data as a data frame (name it 'df') in R

"

```
df <- read.table(file = "21F1001107_W1/data/heights.txt", header = TRUE)
```

df

(row)	Mheight	Dheight
1	59.7	55.1
2	58.2	56.5
3	60.6	56
4	60.7	56.8
5	61.8	56
6	55.5	57.9
7	55.4	57.1
8	56.8	57.6
9	57.5	57.2
10	57.3	57.1
11	58.5	57.6
12	59.7	57.1

W1Q4b

"

Find the dimension of the dataframe using R

"

dim(df)

```
r$> # W1Q4b
"
Find the dimension of the dataframe using R
"
dim(df)
[1] "\nFind the dimension of the dataframe using R\n"
[1] 1375    2
```

W1Q4c

"

Using R, find the summary of each of the columns of the above dataframe.

Comment on the dataset based on output obtained.

"

summary(df)

```

r$> # W1Q4c
"
Using R, find the summary of each of the columns of the above dataframe.
Comment on the dataset based on output obtained.
"
summary(df)
[1] "\nUsing R, find the summary of each of the columns of the above dataframe.\nComment on the dataset based
on output obtained.\n"
  Mheight      Dheight
Min.   :55.40   Min.   :55.10
1st Qu.:60.80   1st Qu.:62.00
Median :62.40   Median :63.60
Mean   :62.45   Mean   :63.75
3rd Qu.:63.90   3rd Qu.:65.60
Max.   :70.80   Max.   :73.10

```

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W1Q4d

"

Print the first and last 10 rows of each column.

"

```
print(head(df, 10))
```

```
print(tail(df, 10))
```

```

r$> # W1Q4d
"
Print the first and last 10 rows of each column.
"
print(head(df, 10))
print(tail(df, 10))
[1] "\nPrint the first and last 10 rows of each column.\n"
  Mheight Dheight
1      59.7    55.1
2      58.2    56.5
3      60.6    56.0
4      60.7    56.8
5      61.8    56.0
6      55.5    57.9
7      55.4    57.1
8      56.8    57.6
9      57.5    57.2
10     57.3    57.1
  Mheight Dheight
1366    66.6    70.0
1367    67.2    70.8
1368    67.6    70.0
1369    68.3    70.8
1370    69.5    70.4
1371    69.1    70.1
1372    65.0    71.6
1373    66.3    71.4
1374    70.8    71.0
1375    63.0    73.1

```

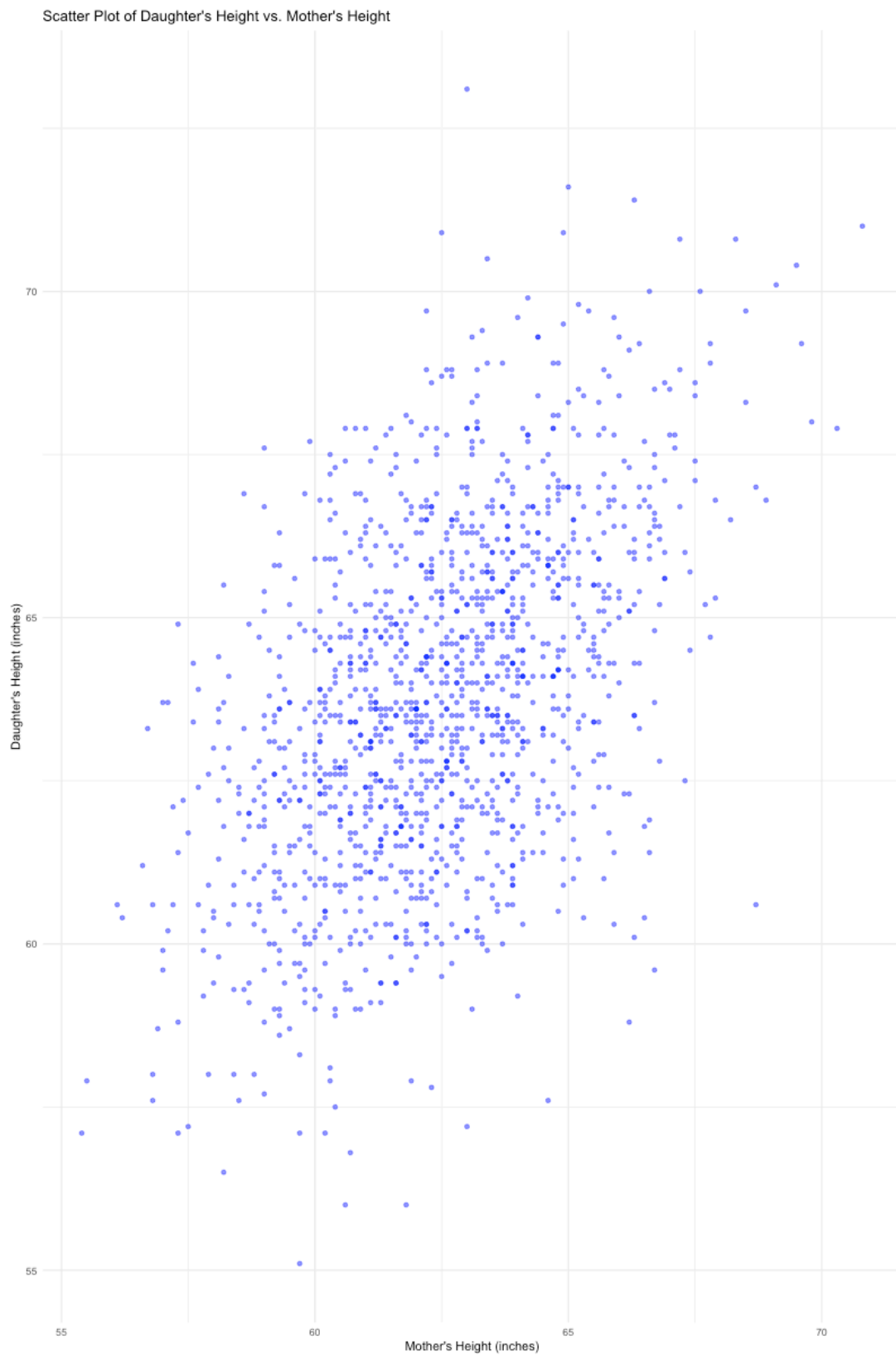
W1Q4e

"

Using the 'ggplot' in R, plot the scatter plot
between the mother and daughter heights.

"

```
ggplot(df, aes(x = Mheight, y = Dheight)) +  
  geom_point(color = "blue", alpha = 0.5) +  
  labs(  
    title = "Scatter Plot of Daughter's Height vs. Mother's Height",  
    x = "Mother's Height (inches)",  
    y = "Daughter's Height (inches)"  
  ) +  
  theme_minimal()
```



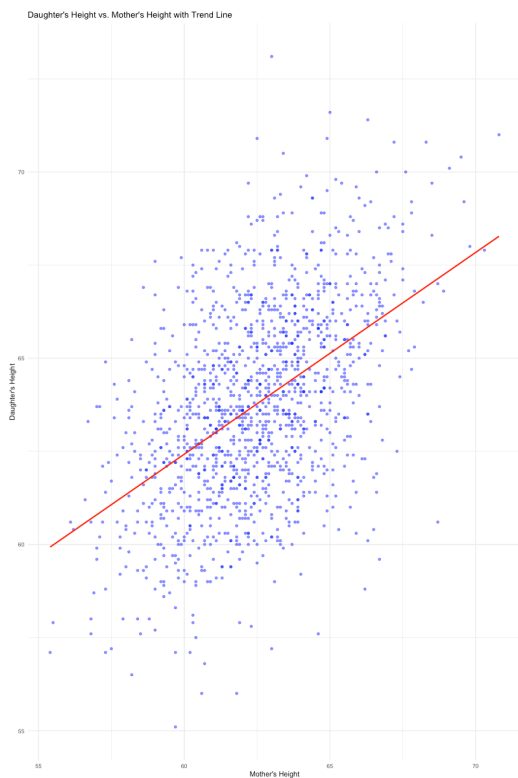

```
# W1Q4f
```

```
"
```

Using 'ggplot()-geoms' in R, draw the function as a continuous curve in the above plotted scatter plot

```
"
```

```
ggplot(df, aes(x = Mheight, y = Dheight)) +  
  geom_point(  
    color = "blue",  
    alpha = 0.5  
  ) +  
  geom_smooth(  
    method = "lm",  
    se = FALSE,  
    color = "red"  
  ) +  
  labs(  
    title = "Daughter's Height vs. Mother's Height with Trend Line",  
    x = "Mother's Height",  
    y = "Daughter's Height"  
  ) +  
  theme_minimal()
```



```
# W1Q4g
```

```
"
```

On visualizing the scatter plot, comment on the independence of the two variables (Dheight and Mheight)

```
"
```

```
ans_w1q4g <- "
```

There is definitely a linear relationship b/w Daughter's and Mother's height, i.e., Dheight and Mheight have a positive linear relationship"